

Week 2 Outline

Sunday, May 11, 2025 8:35 PM

Lesson Plan Template

Gradient of Scalar Function

Scalar Functions vs. Vector Fields

Review and Bridge-In

Learning Objective(s) By the end of this lesson, students will have demonstrated the ability to: Compute gradient, understand gradient as direction of max change, surface normals

Teaching Points and Content Input Strategies (Teaching)	Learning Activities (Application)
Consider the most efficient and active ways possible of delivering content. Gradient of a scalar function Directional Derivative Gradient as direction of max. change Normal to level surface 	Use strategies that get learners interacting with or applying content and skills. Calculate $\vec{\nabla} f(x, y)$ $\vec{\nabla} f(x, y, z)$ Compute $D_{\vec{u}} f = \vec{u} \cdot \vec{\nabla} f$ Compute $\ \vec{\nabla} f\$, $\vec{\nabla} f$, $\ D_{\vec{u}} f\ _{\max}$ Set $f(x, y) = c$, parametrize, compare $\vec{r}'(t)$ with $\vec{\nabla} f$. 

Review and Formative Assessment

Formulate a cone,
Compute $\vec{n} = \frac{1}{\|\vec{\nabla} f\|} \vec{\nabla} f$. Visually inspect $\vec{n}$.

Closure and Link to Outcomes and Summative Evaluation

Common fields that come from scalar potential



Lesson Plan Template

Divergence of a vector field

Review and Bridge-In Scalar vs. vector field

Learning Objective(s) By the end of this lesson, students will have demonstrated the ability to:

Teaching Points and Content Input Strategies (Teaching)	Learning Activities (Application)
Consider the most efficient and active ways possible of delivering content. Definition of divergence Divergence is a scalar The continuity equation Laplacian 	Use strategies that get learners interacting with or applying content and skills. Compute $\vec{\nabla} \cdot \vec{v}$ Foreshadow into changing coordinates and that calculation formula may change, but result remains the same Meaning of divergence $\vec{\nabla}^2 f = \vec{\nabla} \cdot \vec{\nabla} f$ 

Review and Formative Assessment Compute $\nabla^2 f$, $f = e^{xyz}$

Closure and Link to Outcomes and Summative Evaluation Think of some applications of divergence

